Worksheet 4

Instructions

- Include your full name and Coyote ID on the top right of the first page.
- Submit completed worksheets to Gradescope (no tagging required).
- Submit before 11:59 PM on Wednesday, September 30, 2026

Physics Problem Solving Tips

- Draw a diagram (if possible).
- Write down what you know.
- Write down what you want to find.
- First solve symbolically; plug in values at the end.
- Check to make sure your units are correct.
- If you plug in values, check order of magnitude.

Problems

This worksheet covers kinematics in 2D — circular motion..

1. Your roommate is working on his bicycle and has the bike up-side-down. He spends the $60\,\text{cm}$ diameter wheel, and you notice that a pebble stuck in the tread goes by three times every second. What are the pebble's speed and acceleration?
2. A bicycle wheel is rotating at $50\,\text{rpm}$ when the cyclist begins to pedal harder, giving the wheel a constant angular acceleration of $0.50\,\text{rads/s}^2$.
   a. What is the wheel's angular velocity, in rpm, $10\,\text{s}$ later?
   b. How many revolutions does the wheel make during this time?
3. Starting from rest, a DVD steadily accelerates to $500\,\text{rpm}$ in $1.0\,\text{s}$, rotates at this angular speed for $3.0\,\text{s}$, then steadily decelerates to a halt in $2.0\,\text{s}$. How many revolutions does it make?
4. A ball is thrown toward a cliff of height h with a speed of $30\,\text{m/s}$ and an angle of 60° above horizontal. It lands on the edge of the cliff $4.0\,\text{s}$ later.
   a. How high is the cliff?
   b. What is the maximum height of the ball?
   c. What is the ball's impact speed?